KARASIEWICZ Stéphane

Senior Data Scientist

27 September 1989

French/British

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About me

Senior Data Scientist with 7+ years delivering ML solutions from research to production. Specialized in environmental analytics, time-series modeling, and scalable pipelines. Published author of 2 CRAN packages and 6 peer-reviewed publications. Expert in Python/R and translating complex problems into actionable insights.

Skills

- Python (Expert)
- R (Expert)
- SQL
- Bash/Shell scripting
- scikit-learn
- TensorFlow
- XGBoost
- PyTorch
- Supervised/unsupervised learning
- Time-series forecasting
- Clustering
- Pandas
- NumPy
- Dask
- ETL pipelines
- Data lake architecture
- NetCDF/Zarr

Experiences

02/24-01/26 **Senior Data Scientist** — Home based

Raincoat LLC

- ML infrastructure for peril risk assessment
- Engineered end-to-end ML pipeline
- Deployed 5 interactive dashboards with country level maps
- Automated data management system for 2TB+ satellite data
- Analysis reports translating model outputs to insights
- Tools: Python, scikit-learn, Docker, PostgreSQL

07/23-02/24 **Freelance Data Scientist** — Home based

Self-employed - Delivered 12+ ML projects

- Customer churn model (XGBoost)
- Price forecasting (LSTM)
- Custom analytics solutions with Python and R
- Full ML lifecycle: preprocessing, feature engineering, tuning, deployment
- Tools: Python, scikit-learn, TensorFlow, SQL

08/22-07/23 **Data Scientist Certification** — OpenClassrooms (Online)

400+ hour program: Python, ML deployment, cloud

- 10 projects: regression, classification, clustering, NLP, CV
- Deployed 3 ML models via REST APIs with CI/CD
- Recommendation system handling 100K+ interactions
- Interactive dashboards with Streamlit and Dash
- Projects: Segmentation — CNN — NLP — Docker/AWS

10/21-03/22 **Research Engineer** — IFREMER, Boulogne-sur-mer, France

R&D data science for marine research

- Automated 500GB+ data extraction and preprocessing
- Reduced manual effort by 50 hours/month
- Optimized R scripts: 70% computation time reduction
- Automated Rmarkdown reporting (12 annual reports)
- Mentored 4 researchers: R, Git, reproducible research
- Tools: R, Rmarkdown, SQL, Git, ggplot2

09/20-09/21 **Postdoctoral Researcher** — IFREMER, Boulogne-sur-mer, France

Algal bloom detection with satellite imagery & ML

- Bloom detection algorithm: 10+ years satellite data
- Processed 50,000+ km² coastal waters (MODIS, Sentinel)
- Published BDAlgo R package (CRAN)
- Multivariate analysis: 8 environmental drivers identified
- 2 peer-reviewed publications, 1 dataset
- Tools: R, satellite sensing, spatial stats, ML

09/19-09/20 **Freelance Data Scientist** — Freelance, France

Data analysis & translation for 15+ clients

- Statistical consulting and data visualization
- Custom R tutorials for Excel-to-R transitions
- Script optimization: 40-60% performance gains
- Translated 200+ pages technical docs (ENG-FR)
- Tools: R, Python, statistics, visualization

02/18-07/19 **Postdoctoral Researcher** — IFREMER, Centre BREST, France

Marine ecology with satellite data & stats

- Analyzed 15+ years satellite data (1TB+)
- Phytoplankton community dynamics study
- Multivariate techniques (PCA, niche modeling)
- Identified niches of 20+ harmful algae species
- Mentored 2 Master's students
- 2 first-author papers, 1 dataset
- Tools: R, satellite analysis, multivariate stats

Skills (cont.)

- Satellite data processing
- Docker
- Git
- CI/CD
- Linux system admin
- Dash
- Streamlit
- Plotly
- ggplot2
- Shiny
- API development
- Geospatial analysis
- Satellite imagery processing
- Environmental modeling
- Statistical analysis (multivariate, PCA, spatial-temporal)

Education

2014-2017 Ph.D. in Marine Ecology — University Lille, France

Phytoplankton evolution under global changes

- Novel statistical framework (WitOMI) for niche shifts
- Analyzed 20+ years monitoring data (10M+ records)
- Advanced multivariate techniques
- 3 first-author papers, 2 CRAN packages
- Teaching: Biostatistics (87h), Ecosystemic Ecology (16h), Marine sampling (24h)
- Focus: Multivariate analysis, niche theory, biodiversity, time-series

2012-2014 M.Sc. Marine Ecology — Stockholm University, Sweden

Marine community ecology, ecosystem modeling, statistical analysis

2010-2012 B.Sc. (Open University) — Open University, England

Ecosystem Science, Oceanography, Evolution, Environmental Change

2008-2010 Foundation Degree in Animal Science — Bicton College, England

Animal Welfare, Behaviour and Management

2007 BAC S Option SVT — Lycée Marc Chagall, Reims, France

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Open Source & Publications

BDAlgo - CRAN Package. Oct 2023 — 3,000+ downloads — 12+ citations. R package for bloom detection in time-series. Extracts 22 phenological variables. https://cran.r-project.org/web/packages/BDAlgo/

Subniche - CRAN Package. Jul 2022 — 5,000+ downloads — 15+ citations. Statistical package for ecological niche modeling using OMI framework. https://cran.r-project.org/web/packages/subniche/

WitOMI Tutorial: Documentation for Within Outlying Mean Indexes. https://github.com/KarasiewiczStephan/WitOMI

Karasiewicz S., Lefebvre A. (2022). Environmental impact on harmful species. *J. Marine Science & Eng.*, 10(2):174. https://doi.org/10.3390/jmse10020174

Karasiewicz S., Lefebvre A. (2021). Environmental impact on harmful species: Dataset. SEANOE. https://doi.org/10.17882/85097

Karasiewicz S., Chapelle A., Bacher C., Soudant D. (2020). Harmful algae niche responses. *Harmful Algae*, 98:101785. https://doi.org/10.1016/j.hal.2020.101785

Karasiewicz S., Chapelle A., Bacher C., Soudant D. (2019). Harmful algae niche responses: Dataset. SEANOE. https://doi.org/10.17882/69703

Karasiewicz S., et al. (2018). Realized niche analysis of phytoplankton communities. *Harmful Algae*, 72:16-30. https://doi.org/10.1016/j.hal.2017.12.005

Karasiewicz S., Dolédec S., Lefebvre S. (2017). Within outlying mean indexes. *PeerJ*, 5:e3364. https://doi.org/10.7717/peerj.3364

Languages

French: Native — English: Native — Spanish: Basic